

Lecture 19

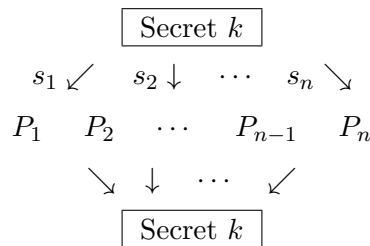
Secret Sharing: Access Structures

SEC 595: Encrypted Computing

Roadmap

Every scheme so far hid a secret by making it hard to compute. Secret sharing hides a secret by *splitting* it, and its security is unconditional. Section 1 states the problem. Section 2 defines access structures, which say which groups may recover the secret. Section 3 defines the dealer as a randomised map, and Section 4 states the two requirements, correctness and privacy. Section 5 gives the first two schemes, XOR and additive sharing. Section 6 lists the main variations.

1. The Problem



Downwards, a dealer computes shares $s_1, \dots, s_n$ and hands s_i to participant P_i . Upwards, a subset of participants pools its shares and reconstructs k . The reconstructing group need not be everybody, so the question is: *which* subsets may recover k , and which must learn nothing?

2. Access Structures

2.1 Collections of Participants

Fix the participants $\{P_1, \dots, P_n\}$. A **collection** is a set of subsets of participants,

$$\mathcal{A} \subseteq 2^{\{P_1, \dots, P_n\}}.$$

Keep the two levels apart: P_1 is a participant, while $\{P_1\}$ is a set of participants and may belong to $\mathcal{A}$.

Definition 1 (Monotone collection). $\mathcal{A}$ is **monotone** if $B \in \mathcal{A}$ and $B \subseteq C$ imply $C \in \mathcal{A}$.

Monotonicity is forced by what reconstruction means: if B can recover the secret, then B plus a newcomer can too, by ignoring the newcomer.

Definition 2 (Minimal sets). $B \in \mathcal{A}$ is **minimal** if no proper subset of B lies in $\mathcal{A}$. The minimal sets of $\mathcal{A}$ form the collection $\mathcal{A}^-$.

Proposition 1. *A monotone collection is determined by its minimal sets: $\mathcal{A} = \{C : B \subseteq C \text{ for some } B \in \mathcal{A}^-\}$.*

Proof. “ $\supseteq$ ” is monotonicity. For “ $\subseteq$ ”, a smallest subset of $C \in \mathcal{A}$ lying in $\mathcal{A}$ is minimal. $\square$

So an access structure is specified by a short list of minimal sets.

Definition 3 (Access structure). An **access structure** is a monotone collection $\mathcal{A}$ of *non-empty* subsets of $\{P_1, \dots, P_n\}$. Sets in $\mathcal{A}$ are **authorized**; sets outside $\mathcal{A}$ are **unauthorized**.

The condition $\emptyset \notin \mathcal{A}$ says that nobody, holding nothing, recovers the secret. Every one of the 2^n subsets is authorized or unauthorized, and the scheme owes something to each: authorized sets *must* recover the secret, and unauthorized sets *must* learn nothing.

Example 1 (Monotone closure). Let $n = 3$ and $\mathcal{A} = \{\{P_1, P_2\}, \{P_1, P_3\}\}$.

(a) *Monotone?* No: $\{P_1, P_2\} \in \mathcal{A}$ and $\{P_1, P_2\} \subseteq \{P_1, P_2, P_3\}$, but $\{P_1, P_2, P_3\} \notin \mathcal{A}$.

(b) *Closure.* The supersets of $\{P_1, P_2\}$ are $\{P_1, P_2\}$ and $\{P_1, P_2, P_3\}$; those of $\{P_1, P_3\}$ are $\{P_1, P_3\}$ and $\{P_1, P_2, P_3\}$. The smallest monotone collection containing $\mathcal{A}$ is

$$\{\{P_1, P_2\}, \{P_1, P_3\}, \{P_1, P_2, P_3\}\},$$

with 3 sets.

(c) *Minimal sets.* $\{P_1, P_2\}$ and $\{P_1, P_3\}$: their proper subsets $\{P_1\}, \{P_2\}, \{P_3\}, \emptyset$ are not in the collection. $\{P_1, P_2, P_3\}$ is not minimal, since it contains $\{P_1, P_2\}$. In words, P_1 together with either other participant can recover.

2.2 Threshold Structures

Definition 4 ((t, n) -threshold). In a (t, n) -threshold scheme any t of the n participants can recover the secret, and no fewer:

$$\mathcal{A}_{t,n} = \{B \subseteq \{P_1, \dots, P_n\} : |B| \geq t\}, \quad 1 \leq t \leq n.$$

Proposition 2. $\mathcal{A}_{t,n}$ is an access structure. Its minimal sets are the $\binom{n}{t}$ sets of size t , and $|\mathcal{A}_{t,n}| = \sum_{i=t}^n \binom{n}{i}$.

Proof. If $B \subseteq C$ then $|C| \geq |B| \geq t$, and $t \geq 1$ excludes $\emptyset$. Counting the authorized sets by size gives the sum. $\square$

The extremes are (n, n) , unanimity, and $(1, n)$, in which everyone can reconstruct alone: replication, with no confidentiality.

2.3 General Access Structures

Not every policy is a count. With participants A, B, C, D , take the policy “only A and B together can recover”. Then $\mathcal{A}^- = \{\{A, B\}\}$, and by Proposition 1 $\mathcal{A}$ consists of the supersets of $\{A, B\}$. No threshold expresses this: $\{A, B\}$ and $\{C, D\}$ have the same size, but only $\{A, B\}$ is authorized.

Example 2 (Counting authorized sets). (a) (n, n) : all n participants are needed; $\binom{n}{n} = 1$ authorized set. $(1, n)$: any single participant suffices; every non-empty set is authorized, $2^n - 1$ sets. $(2, 4)$: any two of four; $\binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 6 + 4 + 1 = 11$ authorized sets. The unauthorized sets are $\emptyset$ and the four singletons, and $11 + 5 = 16 = 2^4$.

(b) $(3, 5)$: $\binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 10 + 5 + 1 = 16$. $(2, 5)$: $\binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 10 + 10 + 5 + 1 = 26 = 2^5 - 1 - 5$.

(c) For “only A and B together”, the authorized sets are

$$\{A, B\}, \{A, B, C\}, \{A, B, D\}, \{A, B, C, D\},$$

and the remaining $16 - 4 = 12$ are unauthorized: $\emptyset, \{A\}, \{B\}, \{C\}, \{D\}, \{A, C\}, \{A, D\}, \{B, C\}, \{B, D\}, \{C, D\}, \{A, C, D\}, \{B, C, D\}$. Suppose $\mathcal{A} = \mathcal{A}_{t,4}$. Since $\{A, B\} \in \mathcal{A}$, $t \leq 2$; but then $\{C, D\} \in \mathcal{A}_{t,4}$, while $\{C, D\} \notin \mathcal{A}$. So no t works.

3. Distribution Schemes

3.1 The Pair $\langle \Pi, \mu \rangle$

Definition 5 (Distribution scheme). A **distribution scheme** is a pair $\Sigma = \langle \Pi, \mu \rangle$, where

$$\Pi : K \times R \longrightarrow K_1 \times K_2 \times \cdots \times K_n$$

and μ is a probability distribution on the set R of random strings.

K is the public **domain of the secret**, and $k \in K$ is the secret. One run of the dealer is $\Pi(k, r) = (s_1, \dots, s_n)$ with $r \sim \mu$; share $s_i \in K_i$ goes to P_i only, and the dealer then discards r . For a set B , write $\Pi(k, r)_B$ for the shares held by the members of B .

3.2 Randomness Is Not Optional

Privacy (Section 4) says two distributions coincide. With no randomness the distributions are single points, so the shares of an unauthorized set must be *constant* in k .

Proposition 3. *If $|R| = 1$ and $\langle \Pi, \mu \rangle$ implements $\mathcal{A}$ with $|K| \geq 2$, then every minimal set of $\mathcal{A}$ is a singleton.*

Proof. Suppose a minimal B contains $P_i \neq P_j$. The sets $B \setminus \{P_i\}$ and $B \setminus \{P_j\}$ are unauthorized, so their shares are constant in k ; together they cover B , so B 's shares are constant and RECON_B cannot return two different secrets. $\square$

In particular, no threshold scheme with $t \geq 2$ is deterministic.

4. Correctness and Privacy

Let K be finite with $|K| \geq 2$. A distribution scheme is a **secret sharing scheme** for $\mathcal{A}$ (it *implements* $\mathcal{A}$) if it is correct and private.

Definition 6 (Correctness). For every $B = \{P_{i_1}, \dots, P_{i_m}\} \in \mathcal{A}$ there is a **reconstruction function** $\text{RECON}_B : K_{i_1} \times \cdots \times K_{i_m} \rightarrow K$ such that

$$\Pr_{r \sim \mu} [\text{RECON}_B(\Pi(k, r)_B) = k] = 1 \quad \text{for all } k \in K.$$

The probability is exactly 1: there is no noise and no failure, unlike the lattice schemes. RECON_B sees only the shares of B , and there is one such function per authorized set.

Definition 7 (Privacy). For every $T \notin \mathcal{A}$, all secrets $a, b \in K$, and every share vector $\langle s_j \rangle_{P_j \in T}$,

$$\Pr_{r \sim \mu} [\Pi(a, r)_T = \langle s_j \rangle] = \Pr_{r \sim \mu} [\Pi(b, r)_T = \langle s_j \rangle].$$

The shares seen by an unauthorized set have the same distribution for every secret: the view of T is independent of k .

Proposition 4. *If the scheme is private and $T \notin \mathcal{A}$, then for any prior on k and any share vector σ_T of positive probability, $\Pr[k \mid \Pi(k, r)_T = \sigma_T] = \Pr[k]$.*

Proof. By privacy $\Pr[\sigma_T \mid k] = c$ for all k , so $\Pr[\sigma_T] = \sum_{k'} c \Pr[k'] = c$, and Bayes' rule gives $\Pr[k \mid \sigma_T] = c \Pr[k] / c = \Pr[k]$. $\square$

4.1 Perfect Privacy

The privacy condition names no adversary, no running time and no hardness assumption: it is **information-theoretic** (perfect) security. In Lectures 6–15, Regev and GSW rest on LWE, El-Gamal on DDH and Paillier on composite residuosity, and the adversary's advantage is negligible. Here it is zero. No parameter grows with λ , and better algorithms or quantum computers change nothing. The guarantee covers the shares only, not a cheating dealer or a forged share. The price is share size.

Proposition 5. *Suppose P_i is non-redundant: some $T \notin \mathcal{A}$ with $P_i \notin T$ has $T \cup \{P_i\} \in \mathcal{A}$. Then $|K_i| \geq |K|$.*

Proof. Fix a share vector σ_T of positive probability; by privacy it occurs for every secret k , with some share $s_i^{(k)}$ for P_i . Correctness of $\text{RECON}_{T \cup \{P_i\}}$ gives $\text{RECON}(\sigma_T, s_i^{(k)}) = k$, so $k \mapsto s_i^{(k)}$ is injective from K into K_i . $\square$

5. First Schemes

5.1 Two-out-of-Two Sharing

Let $K = R = \mathbb{Z}_2$, $n = 2$, μ uniform, and $\Pi(k, r) = (r, k \oplus r)$:

	$r = 0$	$r = 1$
$k = 0$	$(0, 0)$	$(1, 1)$
$k = 1$	$(0, 1)$	$(1, 0)$

Proposition 6. *The XOR scheme implements $\mathcal{A}_{2,2} = \{\{P_1, P_2\}\}$, with $|K_1| = |K_2| = |K|$.*

Proof. $\text{RECON}_{\{P_1, P_2\}}(s_1, s_2) = s_1 \oplus s_2 = r \oplus k \oplus r = k$. In each row of the table, s_1 and s_2 each take 0 and 1 once, so a single share is a uniform bit whatever k is. $\square$

5.2 Additive n -of- n Sharing

Definition 8 (Additive sharing). To share $k \in \mathbb{Z}_p$ among n participants, draw $s_1, \dots, s_{n-1}$ uniformly from $\mathbb{Z}_p$ and set

$$s_n = k - (s_1 + \dots + s_{n-1}) \bmod p.$$

Reconstruction is $\text{RECON}(s_1, \dots, s_n) = s_1 + \dots + s_n \bmod p$.

Proposition 7. *Additive sharing implements $\mathcal{A}_{n,n}$.*

Proof. All n shares sum to k by construction. For any $n-1$ participants and fixed k , the map from $(s_1, \dots, s_{n-1})$ to their shares is a bijection of $\mathbb{Z}_p^{n-1}$ (the missing share is determined by the others and k), so their shares are uniform, each vector having probability $p^{-(n-1)}$ for every k . Smaller unauthorized sets see a marginal of this uniform distribution. $\square$

For $p = 2$ and $n = 2$ it is the XOR scheme.

Example 3 (The XOR scheme). (a) The four cells: $0 \oplus 0 = 0$, $1 \oplus 1 = 0$ in row $k = 0$; $0 \oplus 1 = 1$, $1 \oplus 0 = 1$ in row $k = 1$. So $s_1 \oplus s_2 = k$ everywhere. In each row, $s_1 \in \{0, 1\}$ and $s_2 \in \{0, 1\}$ each appear once, so $\Pr[s_i = 0] = \Pr[s_i = 1] = \frac{1}{2}$ for $k = 0$ and $k = 1$.

(b) Let $\Pr[r = 0] = \frac{3}{4}$. Participant P_2 holds $s_2 = k \oplus r$:

$$\Pr[s_2 = 0 \mid k = 0] = \Pr[r = 0] = \frac{3}{4}, \quad \Pr[s_2 = 0 \mid k = 1] = \Pr[r = 1] = \frac{1}{4}.$$

The distributions differ, so privacy fails for $T = \{P_2\}$.

(c) P_1 holds $s_1 = r$, whose distribution ($\Pr[s_1 = 0] = \frac{3}{4}$) does not involve k at all.

Example 4 (Additive sharing over $\mathbb{Z}_5$). Let $K = \mathbb{Z}_5$, $n = 3$: draw s_1, s_2 uniformly and set $s_3 = k - s_1 - s_2 \bmod 5$. For instance, $k = 3$, $s_1 = 4$, $s_2 = 2$ give $s_3 = 3 - 6 = -3 \equiv 2$.

(a) $\text{RECON}(s_1, s_2, s_3) = s_1 + s_2 + s_3 \bmod 5$; above, $4 + 2 + 2 = 8 \equiv 3 = k$. $\checkmark$

(b) There are 25 equally likely pairs (s_1, s_2) . The pair held by $\{P_1, P_2\}$ is (s_1, s_2) itself, so each value has probability $\frac{1}{25}$. For $\{P_1, P_3\}$, given a target (u, w) there is exactly one choice, $s_1 = u$ and $s_2 = k - u - w$, so the probability is $\frac{1}{25}$; likewise for $\{P_2, P_3\}$. This holds for every k .

(c) If the dealer draws only s_1 and sets $s_2 = 0$, then $s_3 = k - s_1$. Now $\{P_1, P_3\}$ recovers $k = s_1 + s_3$, while s_1 alone and s_3 alone are uniform, and $s_2 = 0$ carries nothing. The structure is $\mathcal{A} = \{\{P_1, P_3\}, \{P_1, P_2, P_3\}\}$, with $\mathcal{A}^- = \{\{P_1, P_3\}\}$; P_2 is redundant.

6. Variations

Variation	What it changes
Threshold	$\mathcal{A}$ depends only on $ B $
General access	$\mathcal{A}$ given by its minimal sets
Multi-span	several schemes share one s_i
Multisecret	one scheme, several secrets
Verifiable	dealer and participants may cheat
Dynamic	participants added, updated, removed
Quantum	shares are quantum states

Each variation is still judged by correctness and privacy. Constructions for threshold and general structures — Shamir, CRT and Brickell — are given in Lecture 20.

Summary

1. An access structure $\mathcal{A}$ is a monotone collection of non-empty sets of participants, determined by its minimal sets. Sets in $\mathcal{A}$ are authorized; the rest are unauthorized.
2. The (t, n) -threshold structure is $\mathcal{A}_{t,n} = \{B : |B| \geq t\}$, with $\sum_{i \geq t} \binom{n}{i}$ authorized sets. General structures, such as “only A and B together”, are not thresholds.
3. A distribution scheme $\langle \Pi, \mu \rangle$ maps $K \times R \rightarrow K_1 \times \cdots \times K_n$; without randomness only singleton minimal sets are possible.
4. Correctness: each authorized B has RECON_B , exact with probability 1. Privacy: each unauthorized T sees the same share distribution for every secret.
5. Privacy is information-theoretic; the price is shares at least as large as the secret. XOR and additive sharing implement $\mathcal{A}_{n,n}$.

Exercises

1. Show that the union and the intersection of two access structures on the same participants are monotone collections of non-empty sets. Show that the complement of $\mathcal{A}$ in the power set is never an access structure.
2. Let $\mathcal{A}^- = \{\{A, B\}, \{C, D\}\}$ on participants A, B, C, D . List $\mathcal{A}$, count its sets, and show it is not a threshold structure.
3. Share $k = 3 \in \mathbb{Z}_7$ additively among three participants with $s_1 = 5, s_2 = 6$. Find s_3 and check reconstruction.
4. Describe what a $(1, n)$ scheme must do and give one. Use Proposition 5 to show $|K_i| \geq |K|$ for every i .
5. Prove the converse of Proposition 3: if every minimal set of $\mathcal{A}$ is a singleton, give a deterministic scheme implementing $\mathcal{A}$ and check both conditions.
6. A participant P_n is *redundant* if $B \in \mathcal{A} \Leftrightarrow B \setminus \{P_n\} \in \mathcal{A}$ for all B . Show that discarding P_n 's share gives a scheme for the induced structure on the other $n - 1$ participants. Why does Proposition 5 say nothing about $|K_n|$? Relate this to Example 4(c).
7. Suppose the two probabilities in the privacy condition differ by at most ε . State an analogue of Proposition 4 for such a scheme.